REPORT ON DATASET: AI LABOR MARKET ANALYSIS & PREDICTIVE MODELING

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ABSTRACT

This report is an analysis of the AI labor market using a dataset of 1,500 job postings collected between 2025 and 2026, with 25 features per record. The study covers many data science aspects: data cleaning and outlier detection, exploratory analysis of salary and skill distributions, dimensionality reduction via PCA, unsupervised clustering of AI Engineering and Governance roles using a K-Means algorithm, and supervised salary estimation through KNN regression. Results confirm that salary is primarily driven by experience level and role category, that the average salary can vary by tens of thousands of euros depending on the month the job was posted, that AI Engineering and Governance form the most clearly distinct clusters in the space defined by the main numerical variables in the database, and that KNN achieves competitive salary prediction accuracy using a combination of numeric and categorical features. Pandas library has been used in all tasks related to loading, cleaning, filtering, and managing all the project's data, and NumPy library has been used for fast mathematical calculations, reshaping arrays, and handling linear algebra. However, other libraries were also used and will be mentioned throughout the report.

Index Terms - AI labor market, K-Means clustering, KNN regression, PCA, salary prediction

1. INTRODUCTION

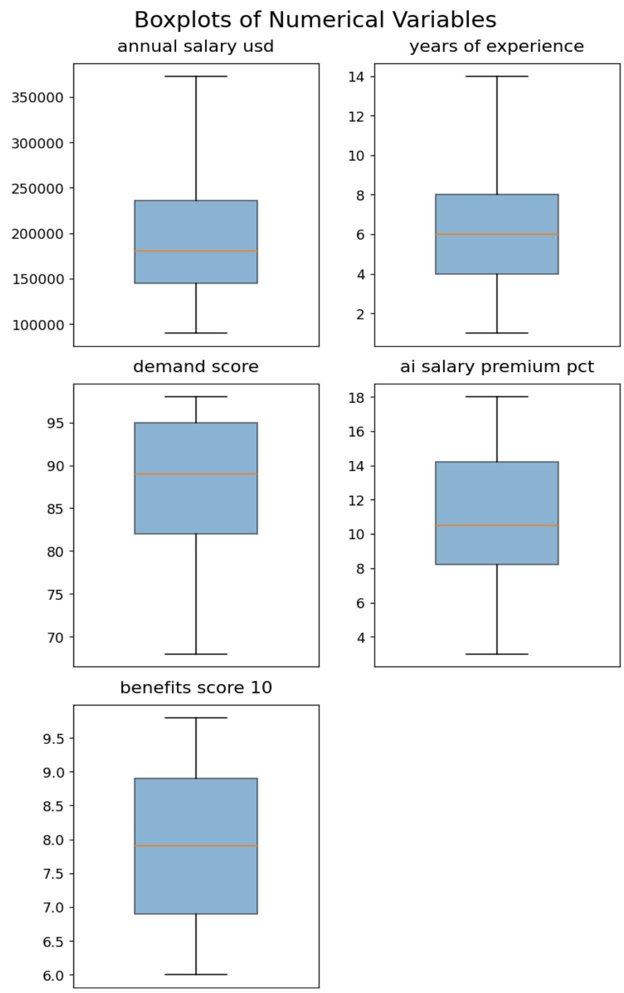
This work analyzes a representative dataset of 1,500 AI-related job postings spanning 2025 and 2026. The dataset includes 25 features covering compensation, experience requirements, remote work arrangements, company size, demanded skills, and demand indicators. The analysis is organized into five methodological stages: data preparation, exploratory data analysis (EDA), principal component analysis, K-Means clustering, and KNN regression for salary estimation.

2. DATA CLEANING AND PREPARATION

The dataset was loaded from a CSV file [1] and verified to be free of missing values. Eight numeric columns were confirmed with correct data types - including annual salary (USD), salary range bounds, years of experience, AI salary premium percentage, demand score, year-on-year demand growth, and benefits score - alongside ten categorical columns covering job title, category, experience level, education, location, remote work policy, company size, industry, and salary tier.

Required skills were parsed into Python lists to enable skill-level analysis. A new column recording the number of skills per posting was also created.

Outlier detection followed the Interquartile Range (IQR) method with a selected threshold of k = 2 and no outliers were detected. Box plots were generated for the five main numeric variables.



3. EXPLORATORY DATA ANALYSIS

3.1. Descriptive Statistics

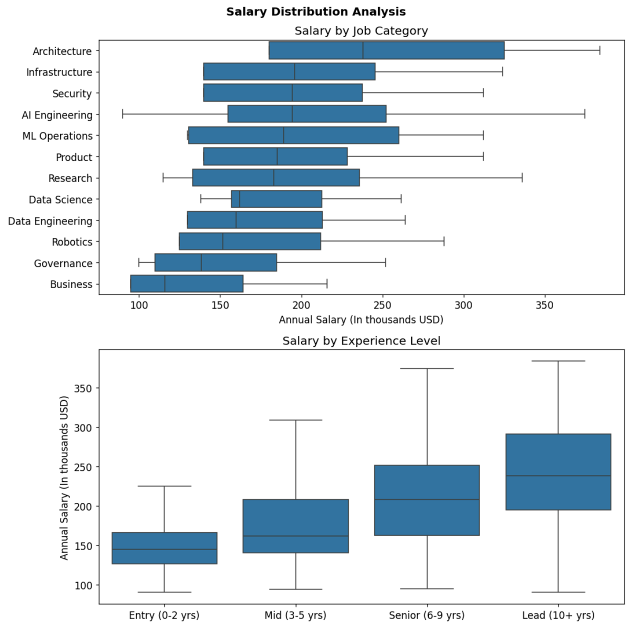
Summary statistics (mean, standard deviation, percentiles at the 10th, 25th, 50th, 75th, and 90th positions, and variance) were computed for all numeric columns. Salary distributions were considerably spread, reflecting genuine heterogeneity across experience levels and role categories.

3.2. Distribution Analysis

Histograms for all eight numeric variables were produced, with mean and median markers overlaid. Salary variables were rescaled to thousands of USD for readability. Pie charts illustrated the distribution of the four main categorical variables: experience level, remote work policy, job category, and company size.

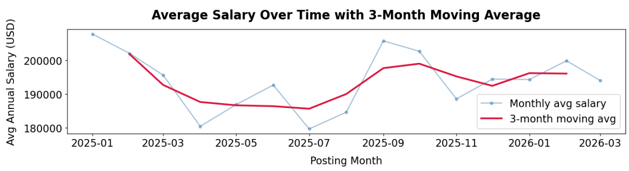
3.3. Salary by Category and Experience

Box plots confirmed a clear positive relationship between experience level and compensation, from Entry (0-2 years) through Lead (10+ years). Sorting job categories by median salary revealed substantial variation between categories.



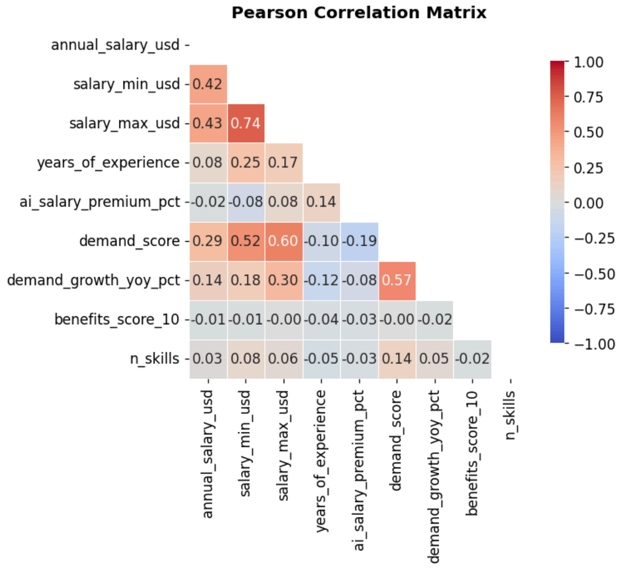
3.4. Temporal Trends

Monthly average salaries were computed from posting year and month fields and visualized as a time series. A 3-month centered moving average was overlaid too.



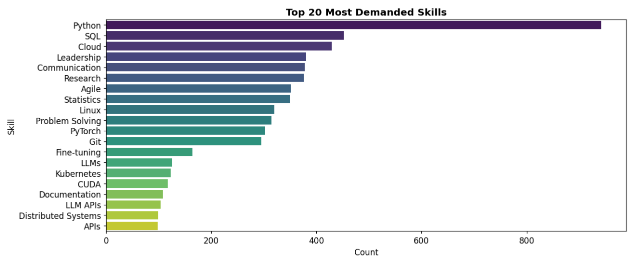
3.5. Correlation Structure

A Pearson correlation matrix was computed for all numeric variables plus skill count. The strongest positive correlations were between maximum and minimum salary, and demand score with maximum salary.



3.6. Skill Frequency

The top 20 most frequently demanded skills were identified by adding all skill lists across all postings. The ranking provides direct insight into which technical competencies are most valued in the current AI job market, with Python winning by more than double to the second skill in position.



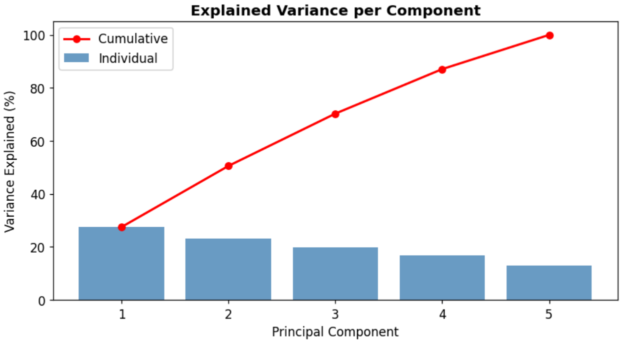
4. PRINCIPAL COMPONENT ANALYSIS

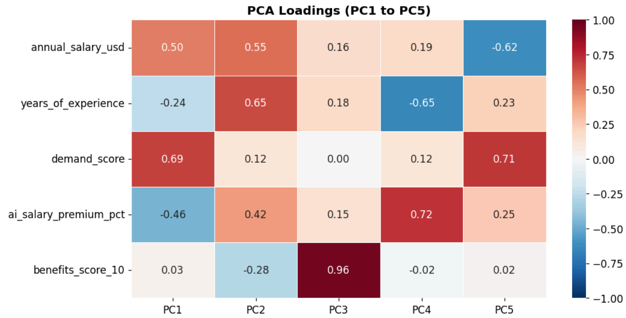
4.1. Feature Selection and Normalization

Five numeric features were selected for PCA: annual salary (USD), years of experience, demand score, AI salary premium percentage, and benefits score. All features were standardized to zero mean and unit variance (Z-scores) using StandardScaler prior to decomposition.

4.2. Explained Variance

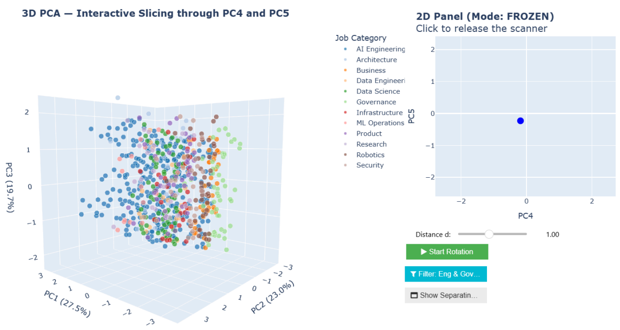
A 5-component PCA was fitted. The first two principal components jointly explained the majority (50.6%) of variance in the feature space. A plot with individual and cumulative variance was produced, and a loadings heatmap for all components was generated to interpret each component's relationship to the original features.





4.3. Interactive 5D Visualization

An interactive Plotly-based 3D scatter plot was built to display job postings in the space defined by PC1, PC2, and PC3. A companion 2D panel allowed users to slice the visualization by PC4 and PC5 values using a hover-based scanner mode and a click-to-freeze locking mechanism. Additional controls enabled category filtering (AI Engineering and Governance only), a distance slider (to determine the range of distance to the subspace of the selected values for PC4 and PC5 in which points appear in the graph), an optional separating plane between cluster centroids, and an auto-rotation feature for exploration. This framework was developed to perfectly visually inspect cluster boundaries and discern which set of job categories had the highest degree of separability between them.



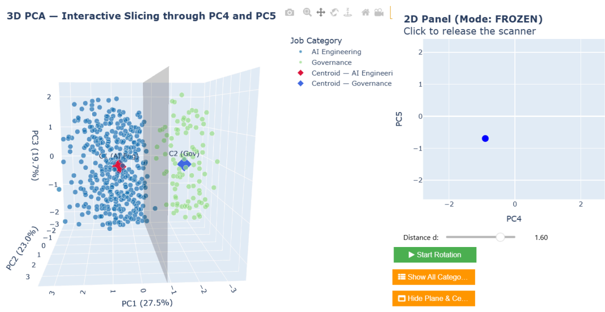
5. K-MEANS CLUSTERING - MANUAL IMPLEMENTATION

5.1. Data Subset

The analysis was restricted to the two job categories that had seemed to present the clearest division in our 5D visualization: AI Engineering and Governance. The PCA-scaled feature matrix was filtered accordingly, yielding a two-class subset.

5.2. Algorithm

A K-Means algorithm with k = 2 was implemented from scratch without the use of scikit-learn's KMeans class. Initial centroids were specified in PCA space and back-projected into the standardized feature space using the inverse PCA transform. The algorithm iterated through the assignment and update steps until the displacement of both centroids fell below a convergence threshold of 0.05. Final centroid coordinates were re-projected into PCA space for visualization in the interactive 5D map.



5.3. Cluster Composition

After convergence, cluster assignments were inspected to verify that the algorithm separated AI Engineering and Governance postings with meaningful purity. The composition of each cluster in terms of true category labels was reported.

5.4. Rand Index

Clustering quality was evaluated using the Rand Index, computed from scratch via pairwise comparison of all point pairs. True positive (TP), true negative (TN), false positive (FP), and false negative (FN) pair counts were tabulated, and the index was calculated as (TP + TN) / (TP + TN + FP + FN). The result , ~0.99, shows an almost perfect agreement between the clustering solution and the ground-truth category labels.

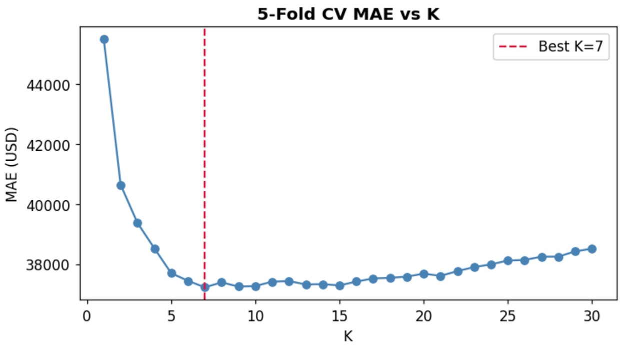
6. KNN REGRESSION - SALARY ESTIMATION

6.1. Feature Engineering

The regression model used eight numeric features (years of experience, demand score, AI premium percentage, benefits score, skill count, and three binary flags for senior, remote-friendly, and LLM-focused roles) plus three label-encoded categorical features (job category, experience level, and company size). All features were standardized using a dedicated StandardScaler before model fitting. The dataset was split 80/20 into training and test sets.

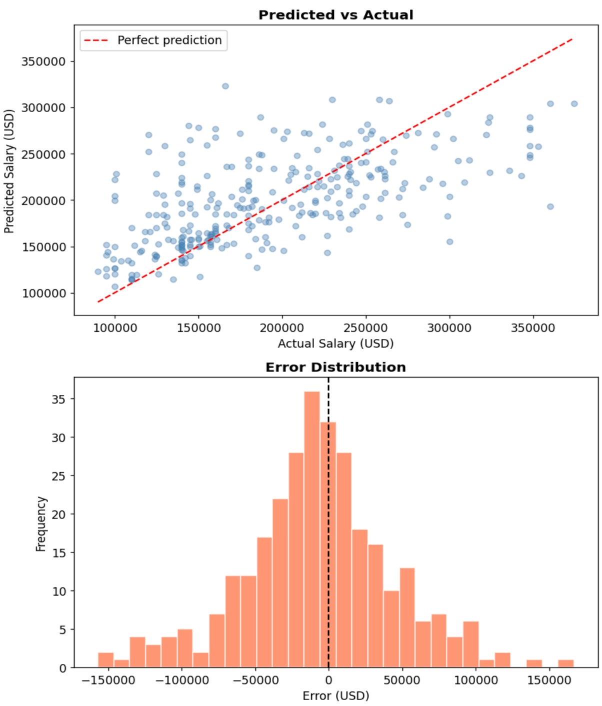
6.2. K Value Selection

Values of k from 1 to 30 were evaluated using 5-fold cross-validation on the training set, with mean absolute error (MAE) as the scoring metric. The optimal k, 7, was identified as the value minimizing cross-validated MAE, and the corresponding CV MAE was reported.



6.3. Model Evaluation

The final KNN model was trained on the full training set and evaluated on the test set, and MAE was reported. Diagnostic plots included a predicted-vs-actual scatter plot with an identity reference line, and an error distribution histogram. The approximately symmetric residual distribution centered near zero indicated absence of systematic bias.



6.4. Prediction Examples

Three illustrative salary estimates were produced for different profiles, which demonstrated the model's ability to differentiate compensation across seniority, category, and company-size dimensions.

7. CONCLUSIONS AND BUSINESS INSIGHTS

This analysis confirms several patterns in the AI labor market. Compensation is strongly differentiated by experience level and job category. PCA effectively optimizes the variance explained components, and K-Means clustering in that reduced space successfully separates AI Engineering from Governance postings with high accuracy as measured by the Rand Index. KNN regression provides a practical and interpretable salary estimation tool, achieving low MAE on unseen data.

Organizations could use the skill frequency analysis to align hiring criteria with market reality, and the salary model to benchmark compensation offers against demand and experience profiles. The PCA visualization further enables exploratory analysis of how different job categories occupy this latent 5-feature space, supporting data-driven planning decisions.

8. REFERENCES

[1] A. Alitaqishah, “AI Jobs Mas Market 2025-2026 | Salaries,” Kaggle, 2026. [Online]. Available: https://www.kaggle.com/datasets/alitaqishah/ai-jobs-market-2025-2026-salaries